
```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;

fm = 500;          % Message frequency
fc = 10000;        % Carrier frequency
fLO = 14550;       % Local oscillator frequency

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% AM Signal
am = (1 + m) .* cos(2*pi*fc*t);

% Local Oscillator
LO = cos(2*pi*fLO*t);

% Frequency Mixing
mixed = am .* LO;

% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed);

% Envelope Detection
env = abs(hilbert(IF));

% Plot Results
figure('Name','AM Superheterodyne Receiver','NumberTitle','off');

subplot(5,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,3);
```

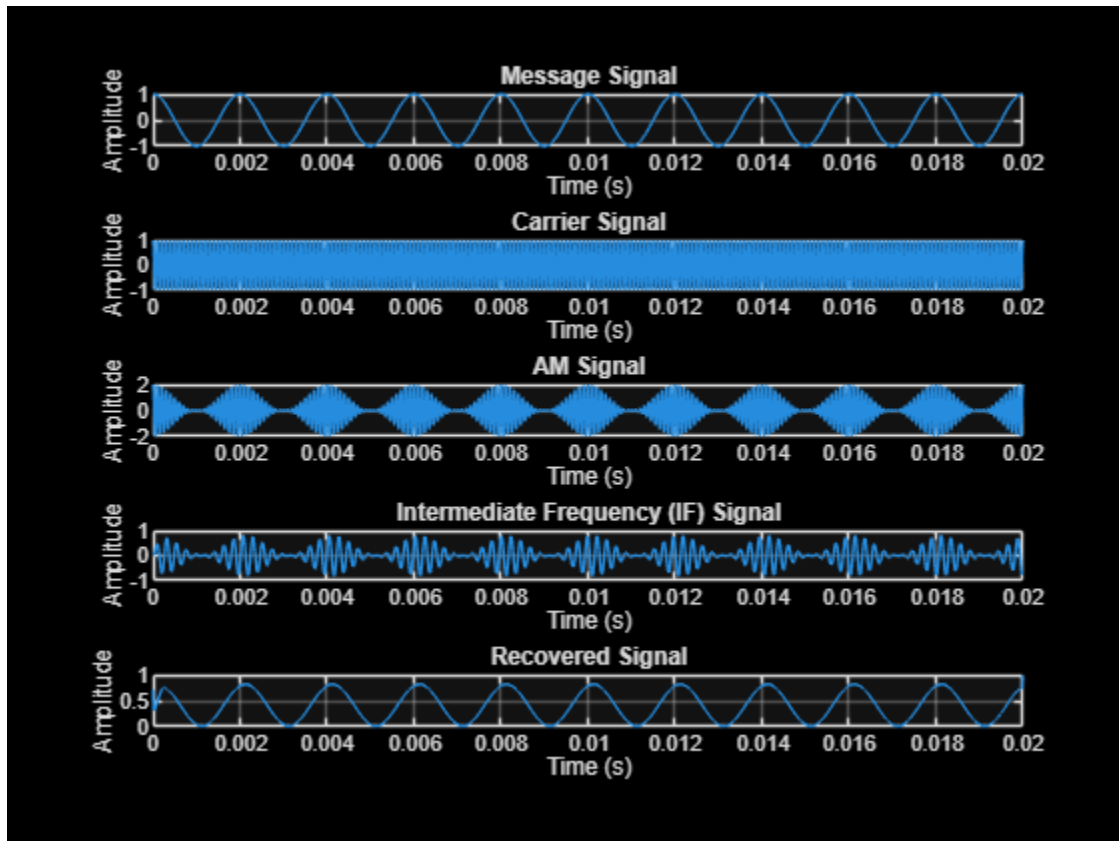
```

plot(t,am);
title('AM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,4);
plot(t,IF);
title('Intermediate Frequency (IF) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,5);
plot(t,env);
title('Recovered Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```



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